pytest-inline: An Inline Testing Tool for Python

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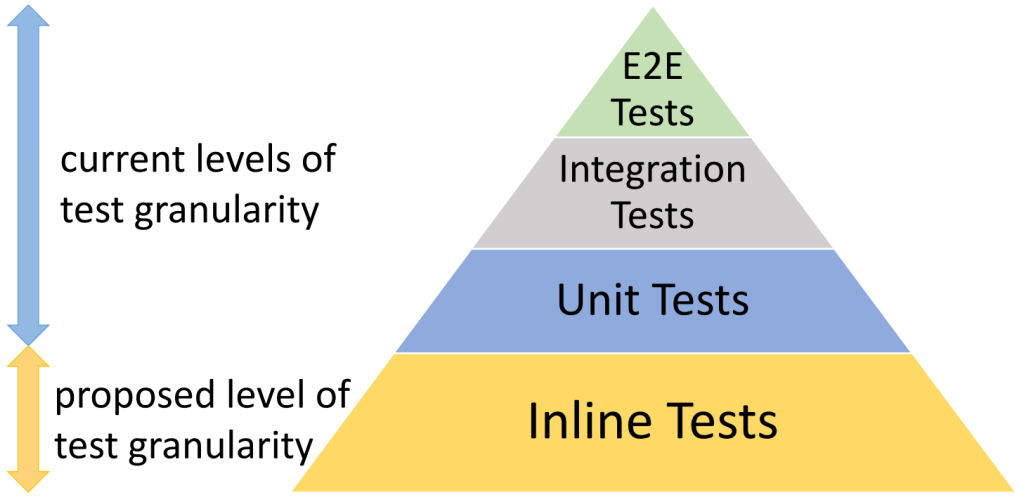
Inline Testing



- Inline testing: new granularity of testing for checking individual program statements
  - regular expression
  - string manipulation
  - bit manipulation
  - collection handling (list comprehension etc.)

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Liu et al., Inline Tests. In ASE 2022.

Inline Testing Example

```
def get_assignment_map_from_checkpoint(tvars, init_c):
  ...
  for var in tvars:
    name = var.name
    m = re.match("^(.*):\\d+$", name)
    itest().given(name, "a:0").check_eq(m.group(1), "a")
  ...
```

statement under test → `m = re.match("^(.*):\\d+$", name)`

inline test → `itest().given(name, "a:0").check_eq(m.group(1), "a")`

declare — assign — assert

Evaluation on 87 inline tests in 31 Python projects

- fast: <0.1s/test
- low overhead: 0.012
- no overhead when disabled

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Liu et al., Inline Tests. In ASE 2022.

Integration with pytest

- `pytest-inline`: a `pytest` plugin that simplifies running inline tests
- `$ pip install pytest-inline`

Feature	Examples
name	`itest(test_name="check_match_name")` `itest("check_match_name")`
parameterized test	`itest(parameterized=True).given(name, ["a:0", "a:1:1"])` `.check_eq(m.group(1), ["a", "a:1"])`
repetition	`itest(repeated=2)`
tag	`itest(tag=["regex"])` `$ pytest --inlinetest-group="regex"`
disabled test	`itest(disabled=True)`

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Integration with `pytest` – New Features

- 5 features added in `pytest-inline` compared to our ASE'22 prototype

Feature	Examples
timeout	`itest(timeout=5)`
assumption	`itest().assume(platform.system() == "Linux")`
execution order	`$ pytest --inlinetest-order="first" --inlinetest-order="second"`
parallel execution	`$ pip install pytest-xdist` `$ pytest -n auto`
more oracles	`check_neq, check_none, check_not_none, check_same, check_not_same`

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Conclusions

- `pytest-inline`: a `pytest` plugin for running inline tests in Python
- Fast, low overhead, easy to write
- Open-sourced and hosted by `pytest-dev` organization

https://github.com/pytest-dev/pytest-inline

```
$ pip install pytest-inline
```

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Backup Slides

A Real Bug Found by Inline Tests

Code from python/cpython, Tools/gdb/libpython.py

```
def write_repr(self, out, visited):
  ...
  elif ch < ‘ ’ or ch == 0x7F:
    out.write(‘\\x’)
    out.write(hexdigits[(ord(ch) >> 4) & 0x000F])
    itest().given(ch, 0x7F).check_eq((ord(ch) >> 4) & 0x000F, 0x07)
    out.write(hexdigits[ord(ch) & 0x000F])
  ...
```

ord(ch)

TypeError: ord() expected string of length 1, but int found